From the definition of $y = \cos^{-1} x$, we observe the following:

(i) $y = \cos^{-1} x$ if and only if $x = \cos y$ for $-1 \le x \le 1$ and $0 \le y \le \pi$.

(ii) $\cos\left(\cos^{-1} x\right) = x$ if $|x| \le 1$ and has no sense if $|x| > 1$.

(iii) $\cos^{-1}\left(\cos x\right) = x$ if $0 \le x \le \pi$, the range of $\cos^{-1} x$. Note that $\cos^{-1}\left(\cos 3\pi\right) = \pi$.

4.4.4 Graph of the inverse cosine function

The inverse cosine function $\cos^{-1} : [-1, 1] \to [0, \pi]$, receives a real number x in the interval $[-1, 1]$ as an input and gives a real number y in the interval $[0, \pi]$ as an output (an angle in radian measure). Let us find some points (x, y) using the equation $y = \cos^{-1} x$ and plot them in the xy-plane. Note that the values of y decrease from π to 0 as x increases from -1 to 1. The inverse cosine function is decreasing and continuous in the domain. By connecting the points by a smooth curve, we get the graph of $y = \cos^{-1} x$ as shown in Fig. 4.14

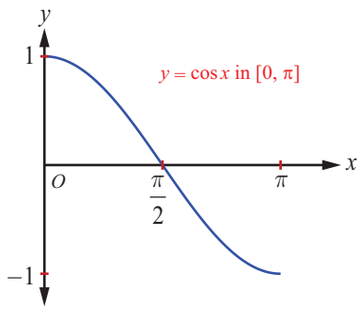


Fig. 4.13

x	y
-1	π
$-\frac{\sqrt{2}}{2}$	$\frac{3\pi}{4}$
0	$\frac{\pi}{2}$
$\frac{\sqrt{2}}{2}$	$\frac{\pi}{4}$
1	0

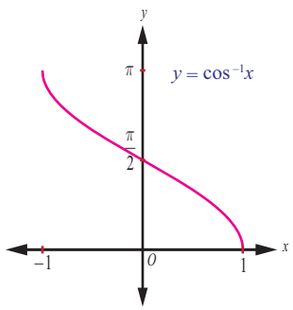


Fig. 4.14

Note

(i) The graph of the function $y = \cos^{-1} x$ is also obtained from the graph $y = \cos x$ by interchanging x and y axes.

(ii) For the function $y = \cos^{-1} x$, the x-intercept is 1 and the y-intercept is $\frac{\pi}{2}$.

(iii) The graph is not symmetric with respect to either origin or y-axis. So, $y = \cos^{-1} x$ is neither even nor odd function.

Example 4.5

Find the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.

Solution

Let $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = y$. Then, $\cos y = \frac{\sqrt{3}}{2}$.

The **range of the principal values** of $y = \cos^{-1} x$ is $[0, \pi]$.

So, let us find y in $[0, \pi]$ such that $\cos y = \frac{\sqrt{3}}{2}$.

But, $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$ and $\frac{\pi}{6} \in [0, \pi]$. Therefore, $y = \frac{\pi}{6}$

Thus, the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is $\frac{\pi}{6}$.

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